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DETAIL SPECIFICATION

MOUNTING KIT: M1 TANK
CLEARED LANE MARKING SYSTEM (CLAMS)

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of a mounting kit used in attaching the cleared lane marking system (CLAMS) to the M1 series tank.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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FEDERAL SPECIFICATIONS

TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-14105	-	Paint, Heat Resisting, For (Steel Surfaces).
MIL-DTL-83413/8	-	Connectors and Assemblies, Electrical, Aircraft Grounding: Type IV Jumper Cable Assembly, Lead, Electrical.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems.
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	-	System Safety Program Requirements.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume 1
MIL-HDBK-505	-	Definitions of Item Levels, Item Exchangeability, Models, and Related Items.

(Copies of the above specifications, standards, and handbooks are available from Document Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or website: https://assist.daps.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

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U.S. ARMY DRAWINGS

13228E2100	-	Mounting Kit, CLAMS to M1 Series Tank.
13228E2030	-	Converter Box Assembly.
13228E2040	-	Cover
13228E2045	-	Panel Assembly, R.H.
13228E2048	-	Panel Assembly, L.H.
13228E2052	-	Panel Assembly, Back.
13228E2066	-	Adapter Assembly.
12344343	-	Interior CARC Paint Specification.
12344344	-	Exterior CARC Paint Specification.

(Copies of these drawings are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of these documents are available from www.asq.org or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications

(Copies of these documents are available from www.asme.org or American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1	-	Structural Welding Code - Steel

(Copies of these documents are available from www.aws.org or American Welding Society, 550 N.W. LeJeune Rd., PO Box 351040, Miami, FL 33135.)

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IPC – ASSOCIATION CONNECTION ELECTRONICS INDUSTRIES

IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

(Copies of these documents are available from www.ipc.org or IPC – Association Connecting Electronics Industries, 3000 Lakeside Drive, 309 S, Bannockburn, IL 60015)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The M1 CLAMS mounting kit provides the interface between the CLAMS and the M1 series tank. The mounting kit shall consist of the following items: a structural adapter for mounting the CLAMS dispenser box at the rear of the vehicle, metallic heat insulation panels that will encase the CLAMS dispenser box and protect its contents from high temperature engine exhaust, a converter box to provide electrical interface between the vehicle and CLAMS control box which controls marker dispensation, and interconnecting electrical harnesses.

3.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the kit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the procuring activity or his designated representative (see 6.2).

3.3 First article (first produced kit). Unless otherwise specified (see 6.2), one or more samples shall be subjected to first article inspection (see 4.3 and 6.3) to prove that his production methods will produce kits that comply with the requirements of this specification. Any changes or deviations of mounting kits from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish mounting kits conforming to this specification.

3.4 Performance characteristics.

3.4.1 Converter box.

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3.4.1.1 Input power. Unless otherwise specified (see 6.2), the requirements specified herein are applicable when an input voltage within the range of 18 to 30 volts direct current (Vdc) is applied to terminals J1-K (positive) and J1-C (return) of the converter box (see 4.5.3.1).

3.4.1.2 Signal processing. For an input sine wave signal of 350 (+35, -0.0) millivolts peak-to-peak (mVp-p) to 4.0 ±1.0 volts peak-to-peak (Vp-p) between terminals J1-E and J1-F at frequencies of 50 ±5.0 to 1000 ±100 Hertz (Hz) the output signal between terminals J1-H (positive) and J1-C (return) shall be a pulse of 17.0 to 30.0 volts maximum amplitude and 0.3 ±0.2 volts minimum amplitude with a pulse duration of 2.0 ±1.0 milliseconds and a signal period of 13.06 (+1.45, -1.19) to 0.653 (+0.073, -0.059) (the converter shall divide the input signal frequency by 653.0) (see 4.5.3.2).

3.4.1.2.1 Noise level. There shall be 0.3 ±0.2 Vdc present between the output terminals J1-H (positive) and J1-C (return) for an input signal between terminals Jl-E and Jl-F of 25 mVp-p or less for frequencies of 5 Hz to 10 KHz (10000 Hz) (see 4.5.3.3).

3.4.2 Structural adapter. The structural adapter shall be assembled in accordance with Drawing 13228E2066. The structural adapter when supported by the telescoping arms, and pintle mounting bar, shall support the mounting of the CLAMS dispenser box and heat insulation panels. The dispenser box shall pivot within the mounting clevis of the structural adapter with all heat insulation panels installed (see 4.5.3.4).

3.5 Design and construction. The mounting kit shall conform to the parts inventory list of 13228E2100.

3.5.1 Dimensions.

3.5.1.1 Structural adapter and heat insulation panels. The CLAMS structural adapter shall be fabricated in accordance with Drawing 13228E2066. The insulation panels shall be fabricated in accordance with Drawings 13228E2040, 13228E2045, 13228E2048, 13228E2052.

3.5.1.2 Converter box housing. The converter box housing shall conform to the requirements of Drawing 13228E2030.

3.5.2 Interchangeability. Mounting kit components shall possess such functional and physical characteristics as to be equivalent in performance, reliability, and maintainability to another item of similar or identical purposes. See MIL-HDBK-505 for guidance.

3.5.3 Soldering. Unless otherwise specified (see 6.2), soldering of electrical connections shall be in accordance with IPC J-STD-001.

3.5.4 Hardware. Mounting hardware shall be selected from AN, NAS, MS, or commercial standards.

3.5.5 Electromagnetic interference/electromagnetic compatibility (EMI/EMC).

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3.5.5.1 Electromagnetic compatibility. The mounting kit, when installed in the vehicle in conjunction with the CLAMS and operated in all its modes of operation, shall not cause adverse effects to the vehicle or any other subsystems installed therein (see 4.5.4).

3.5.6 Bonds and grounds. Electrical bonding for equipment shall be designed in accordance with MIL-STD-464. Equipment mounted electrical connector to mounting surface shall not exceed 50 milliohms. Provisions shall be made for grounding equipment to the vehicle mounting surface. When grounding jumpers are employed, metal bonding strips having a maximum length-to-width ratio of 5 to 1 or braided wire jumpers equivalent to MIL-DTL-83413/8 shall be used. The dc bonding resistance across the enclosure to jumper junction shall not exceed 2.5 milliohms (see 4.5.5).

3.5.7 Reliability. The mounting kit shall have a mission reliability of 0.90 (see 4.5.6).

3.5.8 Maintainability. The mounting kit shall not require preventive maintenance. The mean-time-to-repair (MTTR) at the organizational level of maintenance shall not exceed 1.0 clock hours (see 4.5.7).

3.5.9 Weight. The weight of the mounting kit shall not exceed 413 pounds, including all mounting hardware and kit detail components.

3.5.10 Human factors engineering. The mounting kit shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to requirements of MIL-STD-1472 for labeling, anthropometry (or physical accommodation), design for maintainer, hazards and safety, and general requirements, as applicable.

3.6 Environmental requirements.

3.6.1 Temperature.

3.6.1.1 Operating temperature. The mounting kit shall meet the requirements of 3.4 during and after exposure to temperatures in the range of -25 to +140 ºF (see 4.5.8.1 and 4.5.8.2).

3.6.1.2 Storage temperature. The mounting kit shall meet the requirements of 3.4 after exposure to temperatures in the range of -60 to +160 °F (see 4.5.8.3).

3.6.2 Vibration. The mounting kit shall meet the requirements of 3.4 during and after exposure to sinusoidal vibration in accordance with table I, in each of the three mutually perpendicular axes. The vibration shall consist of a logarithmic sweep rate of 15 minutes per sweep cycle from 5 to 500 to 5 hertz (Hz) (see 4.5.9).

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TABLE I. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Verticle	5 to 25	±1.0 g
	25 to 57	0.03 inch D.A. 1/
	44 to 500	±5.0 g
Longitudinal	5 to 25	±1.0 g
And	25 to 57	0.03 inch D.A. 1/
Longitudinal	44 to 500	±3.0 g

NOTE: 1/ Double amplitude (D.A.)

3.6.3 Basic shock. The CLAMS mounting kit shall meet the requirements of 3.4 during and after exposure to shock impulses of 30 ±3 g, 11 ±1.1 milliseconds (ms) half sine wave applied in each direction of the three mutually perpendicular axes (see 4.5.10).

3.6.4 Ballistic shock. The mounting kit shall meet the requirements of 3.4 during and after exposure to shock impulses of 200 ±20 g, 0.5 ±0.1 ms half sine wave applied in each direction of the three mutually perpendicular axes (see 4.5.11).

3.6.5 High intensity shock. The converter box and parts thereof, shall remain intact and shall not become secondary projectiles when subjected to high intensity shock levels of 980 ±98 g, 0.5 ±0.1 ms at the assembly's mounting interface. This shall include the impulses in each direction of the three mutually perpendicular axes (see 4.5.12).

3.6.6 Submergence. There shall be no evidence of air leakage when the converter box is pressurized with air to 6.0 ±0.5 psig and completely submerged in water to a depth of 6 inches minimum measured from top surface. Electrical connectors shall be capped during submergence. The CLAMS converter box shall meet the requirements of 3.4 after submergence (see 4.5.13).

3.6.7 Cleaning spray. The converter box shall not be damaged by and shall meet the performance requirements of 3.4 after exposure to a jet spray of tap water applied perpendicular to and at a distance of one (+1.0, -0.0) foot from the component surface at a cleaning rate of 1.0 square foot per minute for a period of 10 minutes minimum. The water jet shall be derived from nozzle having a orifice diameter of 0.25 inch and a nozzle pressure of 25 psig (see 4.5.14).

3.7 Treatment and painting.

3.7.1 External. The portions of the mounting kit, not including the deflector, normally painted shall be cleaned, treated, and painted in accordance with Drawing 12344344 color 383 green. The deflector shall be cleaned per applicable method of specification TT-C-490 and painted with MIL-P-14105 color black.

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3.7.2 Internal. The portions of the mounting kit, internally located on the tank, normally painted, shall be cleaned, treated, and painted in accordance with Drawing 12344343 color 17925 untinted white.

3.8 Identification and markings. The mounting kit shall be identified and marked in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall be such that the assembly shall be free of burrs, sharp edges, cracks, corrosion, surfaces out of alignment, or out of contour (visually apparent). There shall be no evidence of conditions that could present a safety hazard to operating or maintenance personnel. The assembly shall contain no loose or foreign material. See MIL-HDBK-454, guideline 9, for guidance on workmanship.

3.9.1 Welding certification.

3.9.1.1 Welding. The surface of parts to be welded shall be free from rust, corrosion, scale, paint, grease, and mill scale that can be removed by chipping and wire brushing, as well as from other foreign matter. Welds shall withstand stress without permanent deformation, or failure, when the parts connected by the welds are subjected to proof and servicing loading. Parts to be joined by fillet welds shall be bought into close contact as possible, and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. The welding process used in fabrication of the mounting kit shall be as specified in applicable drawings. No deviations from these requirements are permissible without prior approval from the procuring activity.

3.9.1.2 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall make available to the contracting officer certification that the welder or welding operator has passed qualification tests as prescribed, by either of the following listed codes, for the type of welding operations to be performed, and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code - Steel
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications

Contractors who make only horizontal welds need not qualify welders or welding operators for "all position welding." The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes. The Government reserves the right to require recertification of any welder or welding operator.

3.9.1.3 Inspection. Inspect all welds as per drawings. No cracks are permissible. Workmanship specimens that are required by the applicable specifications shall be approved by quality assurance representatives of the procuring activity (see 6.2).

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3.10 Safety. The mounting kit shall be designed in accordance with the design guidelines of MIL-STD-882. Guidance for the safety of electrical components can be found in guideline 1 of MIL-HDBK-454.

3.11 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.11.1 Material deterioration prevention and control. The mounting kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.11.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.11.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with each component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.11.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the mounting kit may be newly fabricated from recovered materials to the maximum extent practicable, provided the mounting kit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mounting kit.

3.12 Qualification. Qualification is required for new vendors and shall be performed in accordance with the first article inspection of 3.3 unless otherwise specified by the procuring activity (see 6.2).

4. VERIFICATION

4.1 Reserved.

4.2 Classification of inspection. The inspection specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

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4.3.1 Examination. The mounting kit shall be examined as specified in table II in the order shown. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The mounting kit shall be subjected to tests marked "X" in column 2 of the table III, in the order shown. Failure of any test shall be cause of rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with ANSI/ASQ-Z1.4.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5. Acceptable quality levels (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Examination procedure.

4.5.1 Examination. The mounting kit shall be examined as specified herein for the defects found in table II.

4.5.2 Tests.

4.5.2.1 Test schedule.

4.5.3 Performance characteristics.

4.5.3.1 Input power. Unless otherwise specified (see 6.2), perform the tests specified herein as 18.0 (+0.5, -0.0) Vdc and again at 30.0 (+0.0, -0.5) Vdc. Apply input power to the terminals specified in 3.4.1.1 and verify assembly performance.

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TABLE II. Examination schedule.

First Article	Quality Conformance	Characteristics	Requirement Paragraph
		MAJOR	
X	X	101. Material not as specified	3.2
X	X	102. Components missing, not as specified, or missing.	3.2
X	X	103. Dimensions not as specified.	3.2
X	X	104. Weight not as specified.	3.5.9
X	X	105. Interchangeability not as specified.	3.5.2
X	X	106. Soldering not as specified.	3.5.3
X	X	107. Mounting hardware not as specified.	3.5.4
X	X	108. Human performance/human engineering not as specified.	3.5.10
X	X	109. Welding not as specified.	3.9.1.1
X	X	110. Certification of welders not as specified.	3.9.1.2
X	X	111. Safety not as specified.	3.10
X	X	112. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments as specified.	3.11.1
X	X	113. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other as specified.	3.11.2
X	X	114. Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.11.3
X	X	115. Used, rebuilt, or remanufactured components, pieces, or parts are incorporated in the mounting kit.	3.11.4
		MINOR	
X	X	201. Treatment and painting not as specified.	3.7
X	X	202. Identification and marking not as specified.	3.8
X	X	203. Workmanship not as specified	3.9

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TABLE III. Test schedule.

Test Order	First Article	Quality Conform	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
1	X	X	Input power	4.5.3.1	3.4.1.1
2	X	X	Signal processing	4.5.3.2	3.4.1.2
3	X	X	Noise level	4.5.3.3	3.4.1.2.1
4	X	X	Structural adapter	4.5.3.4	3.4.2
5	X	X	Electromagnetic compatibility	4.5.4	3.5.5.1
6	X	X	Bonds and grounds	4.5.5	3.5.6
7	X		Reliability	4.5.6	3.5.7
8	X		Maintainability	4.5.7	3.5.8
9	X		Low temperature	4.5.8.1	3.6.1.1
10	X	X	High temperature	4.5.8.2	3.6.1.1
11	X		Storage temperature	4.5.8.3	3.6.1.2
12	X		Vibration	4.5.9	3.6.2
13	X		Basic shock	4.5.10	3.6.3
14	X		Ballistic shock	4.5.11	3.6.4
15	X		High intensity shock	4.5.12	3.6.5
16	X	X	Submergence	4.5.13	3.6.6
17	X		Cleaning spray	4.5.14	3.6.7

4.5.3.2 Signal processing. As specified in 3.4.1.2, apply an input signal of 350 (+35, -0.0) mVp-p at a frequency of 50.0 (±5.0) Hz between J1-E and J1-F. Measure and verify the output at terminals J1-H (positive) and J1-C (return) has the following characteristics:

1. Maximum pulse amplitude of 17.0 to 30.0 Vdc and minimum pulse amplitude of 0.3 ±0.2 Vdc.
2. The duration of the maximum amplitude portion of output signal shall be ±1.0 milliseconds.
3. The period of the output signal shall be 13.06 (+1.45, -1.19) seconds.

Repeat the above test with an input frequency of 1000 ±100 Hz in place of the 50.0 Hz.

NOTE: For an input frequency of 1000 ±100 Hz the period (Item 3) shall be 0.653 (+0.726, -0.594) seconds.

Repeat the complete sequence of tests with an input signal of 5.0 (+0.0, -0.5) Vp-p in place of the 350.0 mVp-p signal.

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4.5.3.3 Noise level. Apply an input signal of 25.0 ±2.5 mVp-p with frequency of 3000 ±300 Hz between J1-E and J1-F. Measure and verify the output voltage at J1-H (positive) and J1-C (return) is in conformance with 3.4.1.2.1.

NOTE: This test is to be performed on the 100 percent acceptance tests. The control test for this section shall address the frequencies of 3.4.1.2.1.

4.5.3.4 Structural adapter. Support the structural adapter by the telescoping hull lifting eye mounting tubes and pintle mounting bar. Verify that the dispenser box pivots within the mounting clevis of the structural adapter with heat insulation panels installed, as specified in 3.4.2.

4.5.4 Electromagnetic compatibility. Compliance to 3.5.5.1 shall be determined by conducting an intra system EMG test on an M1 tank with CLAMS and CLAMS mounting kit installed.

4.5.5 Bonds and grounds. Measure the dc resistance, in accordance with MIL-STD-202, method 303, across each bond ground junction with interfacing harnesses disconnected to verify that the dc resistance is as specified in 3.5.6. Visually verify that the bondings are installed in accordance with 3.5.6. Verify the jumpers are as specified in 3.5.6.

4.5.6 Reliability. The mounting kit shall be tested as follows (see 3.5.7):

4.5.6.1 Test conditions. Unless otherwise specified (see 6.2), the reliability shall be conducted under the following conditions:

Air temperature:	73 ±18 °F
Barometric pressure:	28.5 (+2.0, -3.0) inches of mercury
Relative humidity:	50 ±30 percent

4.5.6.2 Accept and reject criteria. The mounting kit shall complete twenty two (22) missions with zero (0) relevant failures. Failures shall be determined by the Government (see 4.5.6.3 for failure definition and 4.5.6.4 for test procedure).

4.5.6.3 Failure definition and scoring criteria. A failure is defined as any malfunction which the operator cannot remedy in 2 minutes or less and which causes any or all of the following: The CLAMS dispenser not to be held in a position for proper operation; the CLAMS not to be properly operated by the crew inside the tank; the CLAMS dispenser not to be protected from the heat of exhaust; the mounting kit and the CLAMS to be mounted in more than 30 minutes. Any failures attributed to the CLAMS or the M1 tank will not be chargeable to the mounting kit. Pattern failures are considered one failure. Failures will be determined by the Government.

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4.5.6.4 Test procedure. A mission for the mounting kit will be as follows:

a. The M1 tank shall rendezvous with the support vehicles at the beginning of the mission.
b. The mounting kit and the CLAMS shall be installed on the M1 tank within 30 minutes.
c. The M1 tank shall then conduct a mine breach of 100 meters over varying terrains. The CLAMS shall be operated during each mission. The automatic dropping of markers shall be different for each mission so that all possible distances are tested at least twice.
d. After the mine field breach is completed, the CLAMS dispenser shall be shut off, and for the duration the 45 minutes operational mode, which started at the beginning of the mine field breach, the M1 tank shall traverse secondary roads at varying speeds up to 10 mph.
e. The position and orientation at the markers dispensed shall be recorded, the distance of each marker from the centerline of the breaching tank and whether the marker is in an upright position. The distance between each marker shall be recorded as well as the setting of the automatic dispensing. All maintenance actions, and times shall be recorded.

4.5.7 Maintainability. To determine conformance to 3.5.8, the time to repair each incident shall be recorded. See MIL-HDBK-470 for guidance.

4.5.8 Operating temperature.

4.5.8.1 Low temperature. Subject the mounting kit to the low temperature test specified in MIL-STD-810, method 502.5, procedure 1 to verify conformance to 3.6.1.1. Maintain the temperature to -60 °F for a period of 24 hours. At the conclusion of this time, stabilize the assembly at the low extreme temperature specified in 3.6.1.1 and perform the test of 4.5.3. Return the assembly to 73 ±18 °F and perform the tests of 4.5.3.

4.5.8.2 High temperature. Subject the mounting kit to the high temperature test specified in MIL-STD-810, method 501.5, procedure 1 to verify conformance to 3.6.1.1. Maintain the temperature at 160 °F for a period of 48 hours. At the conclusion of this time, stabilize the assembly at the high extreme temperature specified in 3.6.1.1 and perform the tests of 4.5.3. Return the assembly to 73 ±18 °F and perform the tests of 4.5.3.

4.5.8.3 Storage temperature. The mounting kit is to be tested in accordance with the tests of 4.5.8 at the temperature extremes specified in 3.6.1.2 to verify conformance to 3.6.1.2.

4.5.9 Vibration. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 514.6, procedure VIII to verify conformance to 3.6.2. Subject the assembly to a sinusoidal vibration test in accordance with the test level requirements of table I, for a total 180 minutes, in each of three mutually perpendicular axes. The vibration is to be imposed at a logarithmic sweep rate of

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15 minutes per sweep with cycling between 5 to 500 to 5 Hz (during exposure of converter box, monitor output signal to verify that intermittent operation does not occur). At the conclusion of this test, examine the assembly for damage and subject it to the performance test of 4.5.3.

4.5.10 Basic shock. Verify conformance to 3.6.3 with the mounting kit mounted, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.6, procedure I. Apply a total of three half sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. With peak amplitude in accordance with 3.6.3 (during exposure of the converter box), monitor the output signal of the converter box to verify that intermittent operations does not occur. At the conclusion of this test, examine the assembly for damage and subject it to the performance test of 4.5.3.

4.5.11 Ballistic shock. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.6, procedure VI to verify conformance with 3.6.4. Apply to total of three half sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude in accordance with 3.6.4 (during exposure of the converter box), monitor converter box output signal to verify that intermittent operation does not occur). At the conclusion of this test, examine the assembly for damage and subject it to the performance tests of 4.5.3.

4.5.12 High intensity shock. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.6, procedure VI to verify conformance with 3.6.5. A total of three half sine wave shock pulses is to be applied in both directions along each of three mutually prependicular axes for a total of 18 shocks. Peak amplitude is to be in accordance with 3.6.5. During exposure the assembly is to be monitored visually to verify it remains intact and that no parts of the assembly become secondary projectiles.

4.5.13 Submergence. To verify conformance of 3.6.6 perform the submergence test as follows:

Install seal caps to all electrical connectors and pressurize the assembly with air through a pressure adapter to 6.0 ±0.5 psig. Submerge the assembly in a suitably-sized container of clear water to a depth of 6.0 (+1.0, -0.0) inch (measure to verify submergence depth between the water surface and the top surface of the assembly) for a minimum period of 5 minutes. Observe the assembly for leakage as evidenced by air bubbles escaping from the interior of the assembly. Any indication of leakage is cause for rejection. Remove the assembly from the water prior to venting the assembly pressure to zero psig. After exposure to the submergence test, subject the assembly to the performance tests of 4.5.3.

4.5.14 Cleaning spray. Subject the mounting kit to a cleaning spray test as specified in 3.6.7. After exposure to the cleaning spray, the assembly is to be subjected to the performance tests of 4.5.3.

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4.5.15 Safety. The mounting kit shall be evaluated for safety requirements throughout testing as specified herein. Nonconformance to 3.10 shall constitute failure of this test.

4.5.16 Human factors engineering. The mounting kit shall be evaluated by a qualified human factors engineer. Nonconformance to 3.5.10 shall constitute failure of this inspection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M1 CLAMS mounting kit permits the mounting of the CLAMS dispenser box and attachment of CLAMS control box and control cable on the Ml series tank.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. If data is required (see 3.2).
d. Time frame required for submission of first article kit (see 3.3 and 3.12).
e. If input power is other than specified (see 3.4.1.1 and 4.5.3.1)
f. Time frame required for submission of welding workmanship specimens (see 3.9.1.3).
g. Government and contractor inspection responsibilities (see 4.1).
h. If test conditions are other than specified (see 4.5.6.1).
i. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the items should be a first produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

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6.4 Vehicle performance. Installation of the mounting kit and CLAMS on M1 series vehicles may degrade vehicle performance in the followings areas:

1. Electromagnetic interference (EMI).
2. Infrared signature.
3. Nuclear survivability.
4. Mobility, i.e., 60 percent incline in reverse, vehicle power response and range, and vehicle towing capability.
5. Power pack removal.

6.5 Subject term (key word) listing.

Converter Box
Dispenser Box
Metallic Heat Insulation Panel
Structural Adapter

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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